

Mark Leader

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Summary

Aerospace engineer with a PhD in Aerospace Engineering from Georgia Tech, focused on computational modeling, structural and thermal finite-element analysis, and multidisciplinary design optimization. Experience at NASA Glenn Research Center includes structural and thermal analysis, battery pack thermal management, hybrid-electric propulsion modeling, and modernization of NASA's Chemical Equilibrium with Applications software. Strong background in developing validated engineering tools, high-performance analysis workflows, and open-source research software.

Education

Georgia Institute of Technology	PhD, Aerospace Engineering	August 2021
	• Thesis: <i>Stress-Based Topology Optimization for Steady-State and Transient Thermoelastic Design</i>	
The University of Texas at Austin	MS, Aerospace Engineering	May 2018
	BS, Aerospace Engineering	May 2016

Experience

NASA Glenn Research Center

Aerospace Engineer – Cleveland, OH June 2022 – Present

Battery Pack Thermal Analysis and Hybrid-Electric Propulsion:

- Developed a hybrid-electric propulsion analysis model coupling time-varying battery cell-discharge and electric motor simulations, enabling aircraft-level optimization studies for reduced climb fuel consumption.
- Implemented phase-change material constitutive modeling in TACS for transient battery pack finite-element simulations, enabling evaluation of passive thermal management strategies.
- Developed a two-way, time-resolved coupling framework for battery pack shape optimization, integrating transient finite-element heat transfer with cell-discharge models.

Modernization of Chemical Equilibrium with Applications CEA:

- Led modernization of NASA's Chemical Equilibrium with Applications software, refactoring a legacy Fortran codebase into a modern, maintainable scientific software package for rocket, shock, and detonation analysis.
- Led public open-source release preparation for NASA CEA, including validation workflows, interface documentation, packaging improvements, and repository maintenance.
- Maintain NASA's open-source CEA repository, managing public releases, user-facing interfaces, validation updates, and long-term software modernization.
- Developed C, Python, and MATLAB interfaces to the Fortran computational core, enabling CEA integration with modern analysis, optimization, and scripting workflows.
- Optimized the Python interface to execute more than 100,000 equilibrium calculations in under 2 seconds, supporting high-throughput design and optimization workflows.
- Developed analytic derivative capabilities for equilibrium thermodynamic outputs, enabling gradient-based integration of CEA into multidisciplinary design optimization workflows.

Peerless Technologies

Aerospace Engineer (Contractor at NASA GRC) – Cleveland, OH June 2021 – June 2022

- Developed propeller optimization capabilities using GXBeam.jl and automatic differentiation, enabling coupled structural and acoustic analysis of design concepts.
- Conducted transient finite-element heat-conduction analyses to predict battery temperatures during thermal runaway, achieving strong agreement with experimental test data.
- Implemented transient heat-conduction analysis capabilities in the open-source MPhys framework using the TACS finite-element solver.

Selected Publications

- Psenica, C., Fang, L., Zoppelt, S., **Leader, M.K.**, and He, P., A Modular Conjugate Heat Transfer Optimization Framework for Thermal Management of Electric Aircraft. *International Journal of Heat and Mass Transfer*, 259:128345, 2026.
- **Leader, M.K.**, Battery Pack Shape Optimization Using Transient Heat Conduction Coupled with Cell-Discharge Analysis. *AIAA SCITECH 2025 Forum*, Orlando, FL, 2025. doi:10.2514/6.2025-0361. AIAA 2025-0361.
- **Leader, M.K.**, Aretskin-Hariton, E., and Moore, K.T., Multidisciplinary Optimization of a Transonic Truss Braced Wing Aircraft with Hybrid-Electric Propulsion. *AIAA AVIATION Forum and ASCEND 2024*, Las Vegas, NV, 2024. doi:10.2514/6.2024-4294. AIAA 2024-4294.
- **Leader, M.K.**, Lavelle, T.M., Wang, X.J., Dickens, K.W., McTague, M., and Hill, J.P., CEA2022: A Modernization of NASA Glenn's Software CEA (Chemical Equilibrium with Applications). *Thermal and Fluids Analysis Workshop (TFAWS) 2024*, Cleveland, OH, 2024. NASA NTRS.
- **Leader, M.K.** and Kennedy, G.J., Thermoelastic Topology Optimization Using Steady-State and Transient Analysis for Stress and Thermal Performance. *AIAA SciTech 2021 Forum*, Virtual, 2021. doi:10.2514/6.2021-1895. AIAA 2021-1895.
- **Leader, M.K.**, Chin, T.W., and Kennedy, G.J., High-Resolution Topology Optimization with Stress and Natural Frequency Constraints. *AIAA Journal*, 2019. doi:10.2514/1.J057777.
- Chin, T.W., **Leader, M.K.**, and Kennedy, G.J., A Scalable Framework for Large-Scale 3D Multimaterial Topology Optimization with Octree-Based Mesh Adaptation. *Advances in Engineering Software*, 2019. doi:10.1016/j.advengsoft.2019.05.004.

Skills

Methods: Finite-element analysis (FEA), transient heat transfer, structural dynamics, buckling analysis, topology optimization, shape optimization, multidisciplinary design optimization (MDAO), chemical equilibrium analysis, high-performance computing (HPC)

Programming: Python, Fortran, C, C++, MATLAB, Julia, Bash, Perl, Cython

Software and frameworks: OpenMDAO, TACS, git, MPI, Nastran, Femap, SolidWorks, NX CAD